

JY Liu

Solutions Architect | Technical Lead | Senior Full-Stack Engineer

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Summary

- Solutions architect and technical lead with 12+ years shipping web, mobile, cloud, and integration-heavy platforms across fintech, SaaS, banking, consular, and enterprise environments.
- Hands-on across frontend, backend, APIs, DevOps, and cloud infrastructure — consistently taking products from discovery and architecture through production delivery.
- Leverages AI tooling pragmatically across the SDLC — analysis, implementation, prototyping, and quality assurance — while building AI-native products end-to-end and maintaining production-grade technical rigor.

Experience

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| Vailley AI Solution Pte. Ltd. , Senior Web Developer (Technical Lead) | Remote (Singapore)
Jun 2025 – Present |
| <ul style="list-style-type: none">• Leading cross-functional delivery for VFS Global's visa and consular platforms — combining solution architecture with hands-on implementation across the application stack, integration layer, and platform.• Delivered a Consular Dashboard with RBAC and ABAC controls, an ADR portal with structured document submission, ETL-backed operational visibility, a digital corporate visa platform for agents, and SAP Concur to OneVasco integration for VFS Global. | |
| Shift Markets , Senior Software Engineer | Remote (NY, US)
Jan 2023 – Present |
| <ul style="list-style-type: none">• Lead engineer for CXM Direct (worldwide multi-asset trading platform) and CRM initiatives, owning backend delivery with ThinkPHP and frontend execution in TypeScript.• Engineered a Unified Payment Gateway integrating 12 PSPs across 12+ regions and 17+ currencies, unifying fragmented payment workflows into a single scalable infrastructure.• Strengthened system reliability and delivery velocity through architectural cleanup, perf tuning, and production-oriented engineering practices. | |
| InternetBrands , Senior Web Engineer | Remote (California, US)
Apr 2023 – Sept 2024 |
| <ul style="list-style-type: none">• Maintained and improved the Dentrux Hub Forms (Online Form for Dentrux & HOP) Laravel backend — patient form management platform for dental practices, focused on reliability, performance, and scale.• Introduced PHP linting and unit testing standards, growing test coverage from 0% to 90% across the codebase.• Managed Kubernetes deployment with Helm and Rancher for the backend. | |
| MaiDaLv , Senior Full Stack Developer | Remote (Guangzhou, China)
Feb 2025 – Apr 2025 |
| <ul style="list-style-type: none">• Built Spring Boot, UniApp, Vue.js, and Python-based product workflows covering subs, WeChat integration, and IP-risk scan. | |
| Cake Experiential Communications Asia , Solutions Architect | Remote (KL, Malaysia)
Apr 2021 – May 2023 |
| <ul style="list-style-type: none">• Led end-to-end architecture and delivery for Vue.js and Laravel platforms used in regional enterprise and campaign engagements.• Introduced engineering standards — TDD, API governance, and backend scaling practices — for client-critical systems. | |
| Glints , Web Software Engineer | Remote (Singapore)
Jan 2022 – Dec 2022 |
| <ul style="list-style-type: none">• Built full-stack recruiter tooling for Superpowered using Next.js and Koa.js, with emphasis on workflow efficiency and usable internal product design.• Strengthened dashboard responsiveness and release quality through improved testing and deployment automation. | |

InnoLab Global , Senior Information Technology Consultant - Delivered cross-platform products across Angular, Ionic, Flutter, Firebase, and AWS while automating cloud delivery workflows and mentoring junior developers across multiple client engagements. - Shipped the Forest City Super App on a Turborepo + Next.js + Prisma + PostgreSQL + Auth.js stack covering community, events, payments, partner portal, and admin console features for residents and partners. - Built Aegis Shooting Academy (ASA) digital platform end-to-end with NestJS + Prisma backend, Next.js admin dashboard, and React Native/Expo mobile application. - Delivered a corporate website and WooCommerce product catalogue for 3V Malaysia (furniture manufacturer) using Docker Compose with a premium Apple-inspired theme. - Architected and led delivery for AvaEra's compensation platform — modernized legacy stack into pnpm + Turborepo monorepo with Next.js, Prisma, PostgreSQL, and a 15-service reward engine covering direct sponsor, team bonus, pool sharing, profit distribution, and merchant commissions. - Built and shipped HappyAI for Gembira Bina Sdn Bhd, an AI-powered tender document extraction and validation platform — Next.js + FastAPI + SQLite with Groq LLM primary / Ollama fallback, confidence-badged review UI, and deterministic batch ingest pipeline.	Remote (KL, Malaysia) Sept 2018 – Present
Supahands , Full Stack Developer Built full-stack features for Supahands' distributed-team product.	Remote (KL, Malaysia) Feb 2018 – Jun 2018
The Consociation LLP , Information Technology Consultant Led full-stack delivery for products including iHeartCity and Swiiipe using Angular, NestJS, GraphQL, and Firebase with AWS-based DevOps automation.	Remote (Singapore) Sept 2016 – Apr 2020
Buddy Holdings Pte Ltd , Senior Information Technology Consultant - Architected and led delivery of the Buddy System Asia MLM platform serving 2,000+ active members across Singapore, Malaysia, China, and Indonesia — Angular + Laravel + MySQL + Kafka covering hierarchy management, point calculations, point redemption, inventory tracking, and family tree updates at scale. - Built agent-facing UI for real-time tracking of progress, points, and family trees across the multi-country user base.	Remote (Singapore) Sept 2016 – Jan 2024
Fuji Xerox Pte Ltd , Embedded Software Engineer Developed embedded software for printer/scanner hardware platforms with C/C++.	Singapore Feb 2016 – May 2016
Netccentric Pte Ltd , Software Developer Built full-stack web features for digital ad campaigns.	Singapore Oct 2015 – Dec 2015
Nexstream Sdn Bhd , Hybrid App Developer Built hybrid mobile apps for iOS and Android using web technologies.	Kuala Lumpur, Malaysia Jun 2014 – Jul 2015

Selected Projects

Unified Payment Gateway Specialized payment infrastructure unifying 20+ global Payment Service Providers (PSPs) into a single, high-concurrency API layer. - Architected and delivered a unified integration layer for 12 PSPs across 12+ regions and 17+ currencies, handling complex transaction workflows, reconciliation, and automated settlement. - Secured financial operations through strict audit logging, sensitive data isolation, and robust error-recovery mechanisms.	Jan 2023 – Present
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VFS Global Consular Platform

Jun 2025 – Present

Mission-critical visa and consular services platform delivering digital transformation for global diplomatic missions.

- Delivered a high-security Consular Dashboard with granular RBAC/ABAC and integrated SAP Concur for automated expense and travel management, supporting multi-country diplomatic operations.
- Built a digital corporate visa platform enabling agents to manage large-scale applications with real-time tracking and ETL-backed analytics.

Superpowered

Jan 2022 – Dec 2022

Talent platform delivered with scalable web architecture, backend APIs, GraphQL, PostgreSQL, and event-driven workflows.

- Built a responsive Next.js and Koa.js application with GraphQL integration, backend optimization, and scalable recruiter tools.

Dentrix Hub Forms (Online Form for Dentrix & HOP)

Apr 2023 – Sept 2024

Dental practice patient form management platform — Laravel backend with Kubernetes deployment, PHP linting, and unit testing standards.

- Maintained and improved the Laravel backend, strengthening response times and concurrency handling for patient form workflows.
- Introduced PHP linting and unit testing, growing test coverage from 0% to 90%.
- Managed Kubernetes deployment with Helm and Rancher for scalable, reliable ops.

Education

Master of Business Administration University of Malaya, Business Analytics
(Expected Oct 2026)

Kuala Lumpur, Malaysia
Oct 2023 – Oct 2026

Master of Computer Science University of Malaya, Mobile Cloud Computing

Kuala Lumpur, Malaysia
Jan 2011 – Jan 2013

Bachelor of Computer Science Universiti Sains Malaysia, Software Engineering

Malaysia
Jan 2008 – Jan 2011

Skills

Full Stack Engineering: TypeScript, JavaScript, PHP, Python, Java, Laravel, ThinkPHP, React, Next.js, Angular, Vue.js, NestJS, Node.js, Spring Boot, Flutter, Ionic

Architecture, Cloud & DevOps: Solution architecture, AWS, GCP, Firebase, Docker, Kubernetes, Helm, Rancher, CI/CD, deployment automation, integration-heavy platform delivery

Backend, Data & Integration: PostgreSQL, MySQL, Redis, Apache Kafka, GraphQL, REST APIs, microservices, ETL workflows, event-driven and serverless systems

AI-Enabled Delivery & Multi-Agent Systems: Orchestrating LLM-based agents, RAG pipelines, prompt engineering, and custom GPT development to accelerate product delivery.

Languages

English, Malay, Cantonese: Professional

Mandarin: Native